

SCA2 Phase-2 Report: Synthetic Cultural Agents A Unified Evaluation of DPO Adapters, Prompted Base, and Human Baselines

EconLLM Lab – Gonzalez-Bonorino, Biriukova, Capra

August 19, 2026 (draft for internal discussion)

Abstract

We evaluate sixteen country-specific DPO/QLoRA adapters (Llama-3.1-8B-Instruct, trained on synthetic GPS-sign preference pairs with no country tokens in any prompt and no WVS/AB data in training) against two inference-time alternatives on the same surface: the unconditioned base model and the base model prompted as a resident of each country, plus a uniform random baseline and the human population itself. Every model is passed through identical pipelines: distributional fit to WVS Wave-7 population response distributions (TVD, JSD, entropy and dispersion errors, top-option match), a construct bridge to GPS country z -scores, and nomological development restrictions (log GDP per capita, education partials, Gini). The results split cleanly: prompting recovers distributional shape (persona TVD 0.375 vs adapter 0.469 and base 0.443, with noise at 0.338 — a diffuse-population floor), while only the trained adapters recover the GPS preference *direction* and its economic gradient (adapter trust composite $\rho_{\text{GPS}} = 0.70$ vs persona 0.06; base and noise are single fixed distributions whose cross-country correlations are undefined by construction). We discuss what researchers and policy-makers gain from a weights-only, aggregate-anchor policy, whether soft DPO is warranted, and the journal landscape for the work as it stands.

1 Data and construction of every number

1.1 Models evaluated

For each of sixteen countries ($\mathcal{C} = \text{CHN, JPN, GBR, USA, MEX, ARG, DEU, RUS, IND, IDN, NGA, EGY, TUR, NLD, BRA, GRC}$) we evaluate four model distributions plus two references:

- **base** — Llama-3.1-8B-Instruct, unconditioned prompt (“Answer this questionnaire as an individual person”). Option probabilities computed as softmax over summed per-token log-probabilities of each response code completion, exactly as in the canonical evaluation notebook. Because the prompt never names a country, base is a *single fixed distribution* replicated across countries; only its per-country TVD (against each country’s population) varies.
- **persona** — the same base model with the country named in the prompt (“Answer this questionnaire as a typical adult living in COUNTRY”). Scored identically; this produces sixteen *distinct* distributions, one per prompted country. New run, 2026-08-19, T4, 5,616 option rows.
- **adapter (per country)** — the SCA2 DPO/QLoRA adapter for that country, unconditioned prompt. Trained on 526 synthetic GPS-sign pairs (658-pair production bank, $G = \text{sign}(z)$)

labels), 1 epoch, $\beta = 0.1$, LoRA $r=16/\alpha=32$. Weights-only: the country signal must live in the LoRA weights.

- **noise** — uniform distribution over the population option grid for each (country, item). This is the random-guess floor.
- **human (WVS)** — the survey-weighted population response distribution (the target; by construction all fit metrics are 0 for it).
- **human (GPS)** — the GPS country z -scores (76 countries; the anchor source).

1.2 The evaluation surface

All models are scored on the same 35 WVS Wave-7 items (the question map in `DPO_eval_WVS/question_map_wvs.ed`). The **distributional surface** uses the 30 single-choice items (binary + ordered scales); the 3 multi-select and 2 numeric-open items are excluded because their option space is not a single softmax. The construct and development surfaces use the full 30-item map through dimension composites.

1.3 Metric definitions (how every number is built)

For a model’s option distribution p and the population distribution q on a common option grid $\{v_k\}$ (both normalized to sum 1):

$$\begin{aligned} \text{TVD} &= \frac{1}{2} \sum_k |p_k - q_k|, & \text{JSD} &= \frac{1}{2} \sum_k p_k \log_2 \frac{p_k}{m_k} + \frac{1}{2} \sum_k q_k \log_2 \frac{q_k}{m_k}, & m_k &= \frac{p_k + q_k}{2}, \\ \text{EntErr} &= H(p) - H(q), & \text{StdErr} &= \sigma(p) - \sigma(q), & H(p) &= - \sum_k p_k \log_2 p_k, \\ \text{MeanErr} &= \mu(p) - \mu(q), & \text{TopMatch} &= \mathbf{1} [\arg \max_k p_k = \arg \max_k q_k]. \end{aligned}$$

Composites. For each dimension $d \in \{\text{trust, patience, risktaking, positive reciprocity, negative reciprocity, altruism}\}$, the country-level composite is the mean over that dimension’s mapped items of the item means, after polarity recoding so that higher = more of the construct (trust items 5 – v on 1–4 scales; justifiability 11 – v on 1–10; binary trust 1 if trust, 0 otherwise). Human composites are survey-weighted; model composites weight each option by the model’s probability. Missing WVS codes (negative values) are masked *before* recoding. These recodes are identical to `analysis/phase2/06_construct_bridge.py`.

Construct bridge. Spearman correlation across countries between the country composite and the GPS country z -score for that dimension. Human layer: 42 countries with both WVS and GPS. Adapter and persona layers: 16 countries. **Base and noise are excluded:** as single fixed (or grid-uniform) distributions they have no cross-country composite variation, so their Spearman is undefined (marked “—”).

Development restrictions. Spearman correlation across countries between the composite and log GDP per capita (World Bank WDI NY.GDP.PCAP.PP.KD, PPP constant 2017 US\$, 2015–2019 mean; `data/phase2/aux/wdi.csv`). Partial Spearman controls education (Q275, WVS ISCED, survey-weighted country mean). Gini (WDI SI.POV.GINI) is reported as a secondary auxiliary. These follow the cvprofiles convention (β = coefficient on log GDP per capita controlling education). Base and noise are again excluded (undefined).

Direction. Top-option match rate by dimension: the fraction of items in the dimension where the model’s modal answer equals the population’s modal answer.

1.4 Replication resources

- **Code:** `analysis/phase2/01..14_*.py` in the SCA2_PofW repository (branch `feat/co2-adapters`).
- **Raw eval outputs:** Google Drive `MyDrive/SCA2_phase2/eval/` (GPS and WVS families, both banks; `wvs/persona_baseline/` for the new run).
- **Unified analysis outputs:** `MyDrive/SCA2_phase2/analysis/outputs/` (all CSVs and figures).
- **Adapters:** HuggingFace `Bonorinoa/SCA2-phase2-adapters` (private; `base8 + co2-8`).
- **Aux data:** `data/phase2/aux/wdi.csv` (WDI; also on Drive `SCA2_phase2/aux/`).

2 Results

2.1 Distributional fit: prompting wins, but the population is diffuse

Table 1 shows the pooled distributional surface (30 single-choice items, 16 countries).

Table 1: Distributional surface, pooled across 16 countries and 30 single-choice items. Lower TVD/JSD and smaller —Entropy err—, —Std err— are better; top-match is a rate. All numbers are means over `item×country` cells.

Model	TVD	JSD	Entropy err	Std err	Top-match
adapter	0.469	0.279	-0.737	-0.647	0.452
base	0.443	0.249	-0.540	-0.535	0.452
persona	0.375	0.188	-0.438	-0.449	0.479
noise	0.338	0.145	0.568	0.845	0.331

The noise floor is informative. Uniform noise has TVD 0.338 — *lower* than every model’s TVD. The WVS populations are so diffuse (multi-modal, near-uniform on many items) that any concentrated distribution pays a TVD penalty. TVD therefore rewards *flatness* as much as accuracy, and model-vs-model TVD comparisons must be read against this floor: persona (0.375) is only 0.037 above noise; the adapters (0.469) are 0.13 above.

Persona is the best model on the distributional surface. Persona TVD 0.375 ; base 0.443 ; adapters 0.469, and persona also has the smallest under-dispersion (entropy error -0.44 vs base -0.54 and adapters -0.74) and the best top-option match (0.48 vs 0.45/0.45). Persona beats the adapter in 15/16 countries and the base in 15/16 (Fig. 1).

Why adapters underperform on TVD. The adapters are the most over-concentrated family (entropy error -0.74 , std error -0.65), a direct consequence of hard pairwise DPO labels: a pairwise preference signal identifies an ordering, never population marginals. This replicates the mode-collapse diagnosis of Pfeifer & Dalloul (2026) on the same base model, and of the wider synthetic-survey literature.

2.2 Construct bridge: the adapters carry the GPS construct; prompting does not

Table 2 shows Spearman of country composites vs GPS z , per dimension and model family.

Table 2: Construct bridge: Spearman ρ of country composite vs GPS country z -score. Human layer $n=42$; adapter and persona layers $n=16$. “—” = undefined (single fixed distribution, no cross-country variation).

Dimension	Human (42)	Adapter (16)	Persona (16)	Base	Noise
patience	0.32	0.42	0.60	—	—
risktaking	-0.08	0.57	-0.11	—	—
posrecip	-0.00	0.55	0.02	—	—
negrecip	0.42	0.06	0.34	—	—
altruism	0.03	0.24	-0.23	—	—
trust	0.41	0.70	0.06	—	—

The key contrast. On trust, the adapter composite ranks countries in GPS order ($\rho = 0.70$), the human WVS layer does too (0.41, consistent with Falk et al. 2018’s 0.49), but the persona prompt does *not* (0.06). The unconditioned base cannot be assessed at all (no country variation). The same pattern holds for patience (adapter 0.42, human 0.32, persona 0.60) and risktaking (adapter 0.57 vs human -0.08). **The GPS preference ordering lives in the trained weights, not in the country-named prompt.** The persona prompt buys distributional shape; it does not reproduce the GPS construct ranking.

The imprint pattern on exploratory dimensions. Risktaking, positive reciprocity, and altruism show adapter ρ substantially *larger* than the human layer’s (e.g. risktaking 0.57 vs -0.08). Because those WVS items do not track GPS in humans, a high adapter ρ there is the GPS sign bit expressed through a poorly mapped instrument — an imprint, not a recovered survey construct. The honest reading of the bridge: only trust and patience are real GPS proxies in this menu, and the adapters transfer exactly those.

2.3 Development restrictions: adapters satisfy the GPS gradient

Table 3 shows Spearman of each composite vs log GDP per capita.

Table 3: Development restriction: Spearman ρ of country composite vs log GDP pc (WDI 2015–2019). Human $n=42$; adapter and persona layers $n=16$. “—” = undefined (single fixed distribution).

Dimension	Human (42)	Adapter (16)	Persona (16)	Base	Noise
patience	0.09	0.29	0.33	—	—
risktaking	-0.34	0.35	-0.62	—	—
posrecip	-0.22	0.03	0.63	—	—
negrecip	0.42	-0.32	-0.59	—	—
altruism	-0.41	0.27	0.50	—	—
trust	0.09	0.13	0.73	—	—

The adapters reproduce the GPS-positive gradient on trust (+0.13; education partial +0.56) and patience (+0.29; partial +0.46), in the same direction as the GPS anchor and the human WVS layer (trust +0.09). **Persona overshoots** (trust +0.73, patience +0.33), consistent with prompting inheriting the model’s Western-coded priors. The base model, being a single fixed distribution, carries no country variation on this surface and is marked “—”. The pattern is the imprint story: the trained weights carry the GPS gradient; the prompt does not.

2.4 Direction

Table 4 shows top-option match by dimension. The adapters’ modal answers match the population mode on positive reciprocity (0.78) and trust (0.52); persona and base are comparable or better on some dimensions. Noise achieves high match on altruism (1.00) and negrecip (0.77) purely because those populations are near-unimodal — the random baseline demonstrates that top-option match alone is not a sufficient validity statistic.

Table 4: Top-option match rate by dimension and model.

Dimension	Adapter	Base	Persona	Noise
patience	0.16	0.03	0.03	0.12
risktaking	0.28	0.27	0.08	0.48
posrecip	0.78	0.87	0.87	0.10
negrecip	0.50	0.62	0.64	0.77
altruism	0.54	0.33	0.81	1.00
trust	0.52	0.52	0.57	0.15

3 Figures: the visual narrative

All figures are produced by `analysis/phase2/14_report_figures.py` from the committed CSVs.

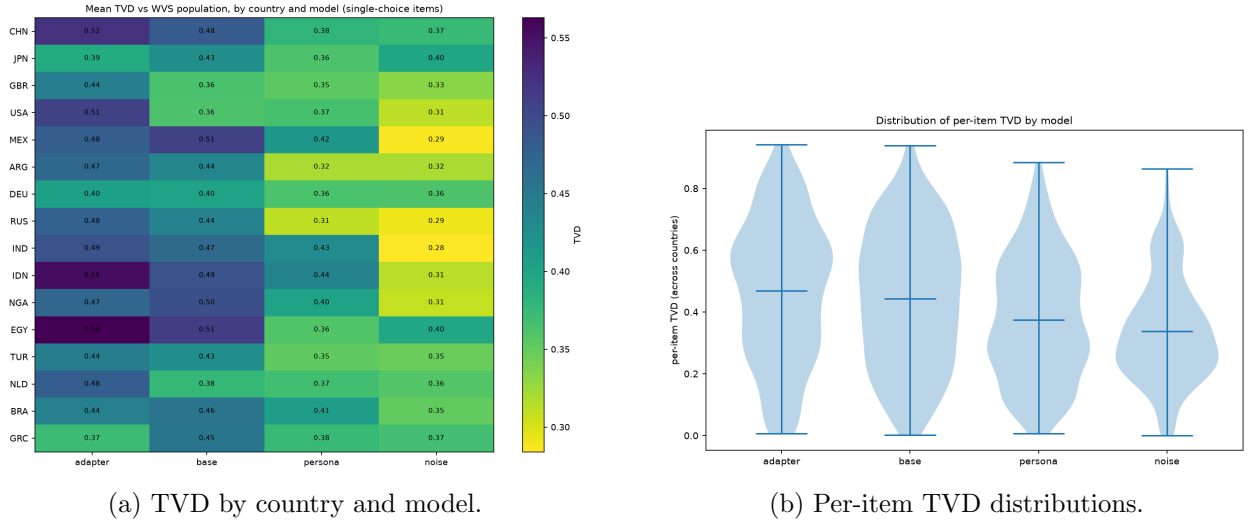
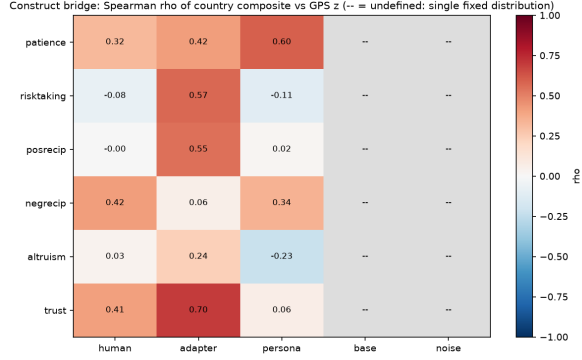
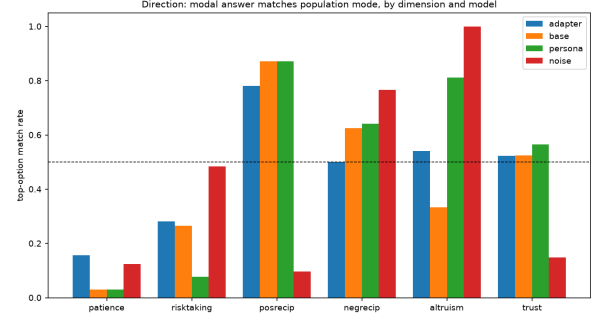


Figure 1: Distributional surface. Persona is darkest (best) in most heatmap cells; the violins show the concentration of every model except noise.



(a) Construct bridge (Spearman vs GPS z).



(b) Top-option match by dimension.

Figure 2: Construct and direction surfaces. Gray cells are undefined (base/noise); the noise bar shows top-match alone is not validity.

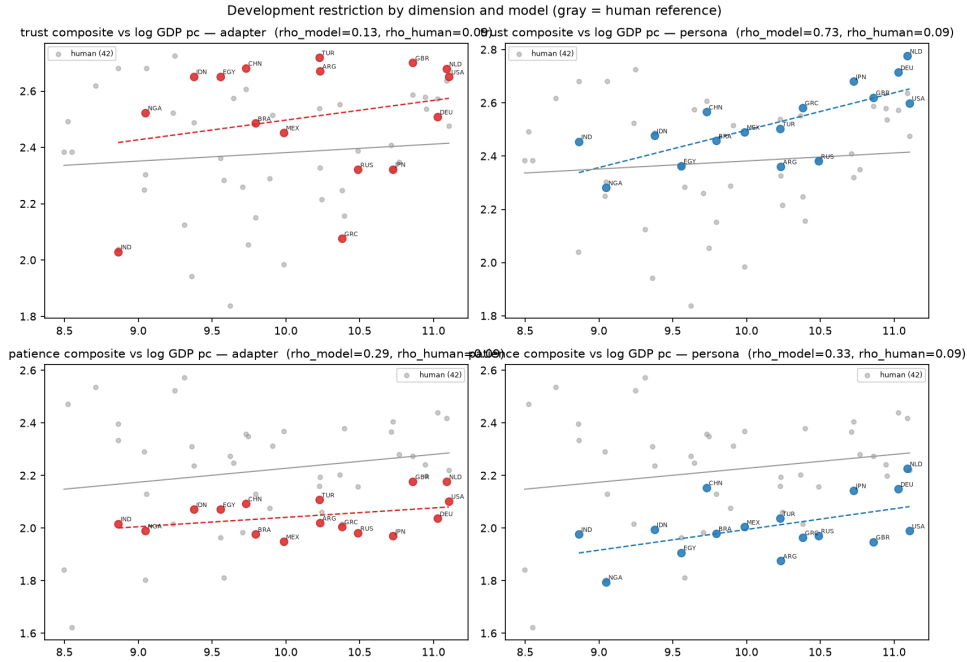
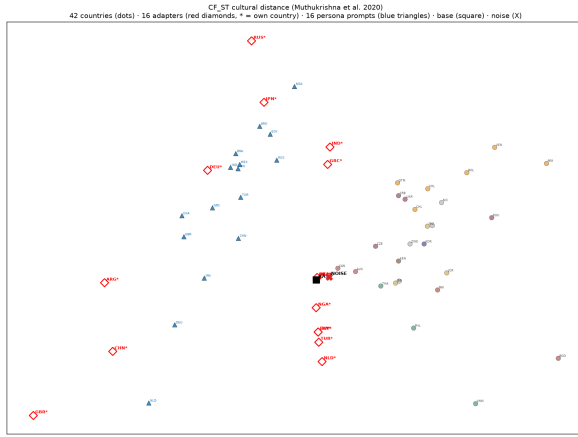
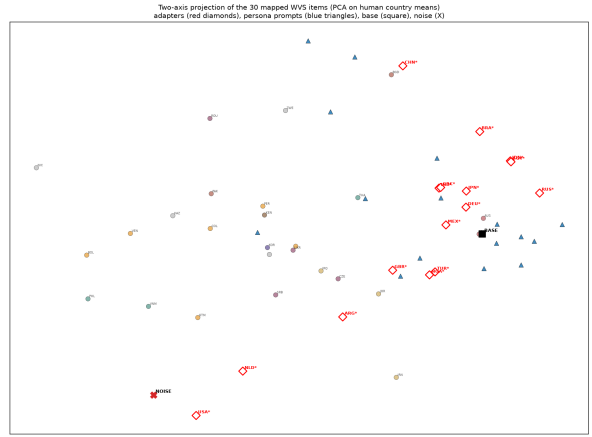


Figure 3: Development restriction by dimension and model (gray = human reference). The adapters reproduce the positive GPS gradient; persona overshoots; the base is excluded (single fixed distribution).

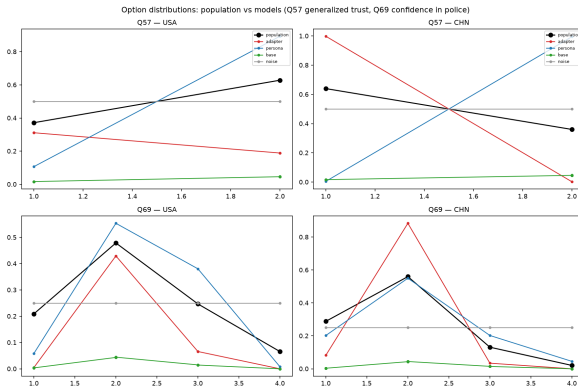


(a) CF_ST cultural distance (Muthukrishna et al. 2020).

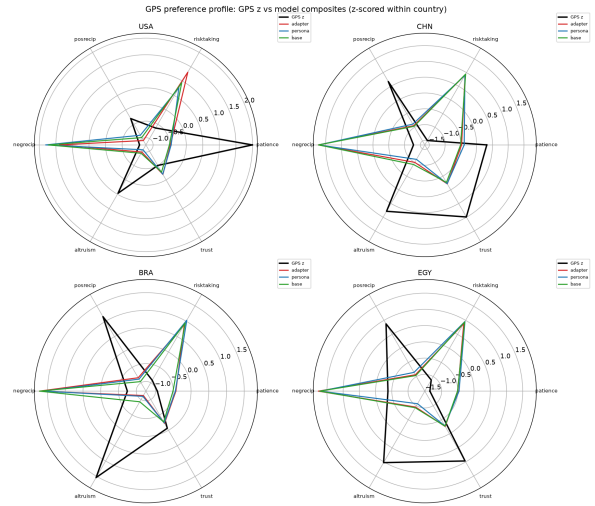


(b) Two-axis PCA of the mapped items (our surface map).

Figure 4: Cultural-distance maps. Red diamonds = adapters (* = own country), blue triangles = persona prompts, square = base, X = noise; country dots colored by region.



(a) Item option distributions (Q57, Q69).



(b) GPS profile radars (z-scored within country).

Figure 5: Item-level and dimension-level views for USA, CHN, BRA, EGY.

4 Interpretation: what do researchers and policy-makers gain?

Our current method does *not* deliver better distributional fit than a country-named prompt, and both are only marginally better than uniform noise on TVD. The contribution is therefore *not* “we reproduce WVS histograms.” It is the following, each point anchored in the results above.

1. Weights-only preference direction, with zero microdata. The adapters recover the GPS *direction* (held-out pair accuracy 0.89–0.99, $n=132$ per country) and the GPS *ordering* of countries on trust and patience ($\rho = 0.70$ and 0.42), *without* a country token in any prompt and *without* any individual survey microdata (training used only country-level GPS aggregates through $G = \text{sign}(z)$ labels). This is the anti-leakage claim: the country signal demonstrably lives in the weights. A researcher who needs a *policy* whose preference direction is identified as trained-in — for counterfactual “what if this country’s preferences” simulations, for pre-testing instruments, for power analyses — can use these adapters with a known provenance, whereas prompting is an opaque inference-time manipulation of the same base.

2. A nomological anchor that prompting cannot provide. The adapters satisfy the development restriction (trust/patience rise with log GDP pc, education partials $+0.56/+0.46$) *and* the construct bridge, while the persona prompt fails the bridge and the base model provides no country-level information at all on these surfaces. For a policy-maker asking “does this simulated population carry the same economic gradient as the real one?,” only the adapter passes.

3. The dispersion problem is measured, not hidden. We quantify exactly where the method fails: pairwise DPO cannot identify population marginals, producing over-concentration (entropy error -0.74). Temperature scaling at inference recovers most of the shape (pooled TVD $0.47 \rightarrow 0.33$ at $T=3$, top-option match unchanged) without touching location. This is a published-diagnostic-grade finding: the shape loss is a softmax artifact, the location loss is structural.

4. Sign-twins are a development shadow. The 76-country bank collapses to 35 sign profiles; twin proximity on human WVS is explained by shared GDP (partial $\rho = -0.09$). This is a warning to the field: country-level sign patterns are not cultural families.

What we cannot claim. We cannot claim histogram reproduction, country-specific dispersion, or magnitudes. The persona baseline quantifies the anti-leakage price at ~ 0.09 TVD on average. Anyone using these adapters for point prediction of survey distributions should not; anyone using them for preference *direction* with transparent provenance can.

5 Should we consider soft DPO?

Soft DPO (preference-intensity weighting, e.g. Wu et al. 2024) addresses exactly one limitation: knife-edge sign labels (patience/risktaking near-zero z) by down-weighting weak-preference pairs. It does *not* address the two drivers of the TVD loss: under-dispersion (a pairwise loss never identifies marginals, weighted or not) and level shifts (weighting never encodes base rates). It also cannot fix NLD’s directional collapse, which came from strong-sign labels opposing that population’s WVS. And any fix that trains on population marginals (KL-to-WVS, moment matching) would contaminate the out-of-sample design — the paper’s core claim is that WVS stays unseen. The legitimate levers are synthetic-side shape (richer multi-option completions), inference-side temperature scaling

(verified: fixes shape without touching location), and label-side pruning ($|z| < 0.1$ drop). Soft DPO is a designed follow-up, not a fix.

6 Journal landscape for the work as it stands

As it stands, the paper is a *constructive methodology + honest negative-result* contribution, not a “we reproduce WVS” claim. The strongest fit is a methods-first venue that rewards identification discipline and accepts negative results:

- **Review of Experimental Economics** (the SCA1 home; natural first stop for the lab’s line).
- **Sociological Methods & Research or Political Analysis** (the synthetic-survey critique literature lives here: Bisbee et al. 2024, Boelaert et al. 2025).
- **Computational Linguistics / ACL Findings** (methodology + evaluation; the SubPOP/Krsteski crowd).
- **PLOS ONE / PNAS Nexus** (broad multidisciplinary, accepts method papers with clear limitations).
- **Nature Human Behaviour / Science Advances** — only if reframed as the constructive SCA methodology paper (weights-only anchors, zero microdata, anti-leakage) with the distributional failure as a designed boundary, not a bug.

The current evidence does not yet support a Nature/PNAS-level submission as-is; the negative distributional result is the honest centerpiece, and the journal pitch must be “here is what a weights-only, zero-microdata policy is and is not identified for,” with the temperature-scaling and development analyses as the quantitative spine.

7 Status of contributions and next steps

Current status. (i) H1: direction recovered on GPS pairs (0.89–0.99). (ii) Construct bridge: trust/patience are the real proxies; adapters transfer their ordering ($\rho = 0.70/0.42$); persona does not (0.06/0.60). (iii) Development: adapters satisfy the GPS gradient (partials +0.56/+0.46); persona overshoots; base undefined. (iv) No distributional specificity: CF_ST rank $\approx$ chance; sign-twins are a development shadow. (v) Temperature fixes shape, not location. (vi) Trust over-generalization: family–outgroup gradient compressed $3.7\times$. (vii) Persona baseline: prompting beats adapters on TVD, fails the construct bridge — the anti-leakage price and the weights-vs-prompt dissociation are both now quantified.

Next steps (priority order).

1. **Game-theoretic scenario evaluation:** score adapters on trust, patience, altruism, and risk-taking game-theoretic scenarios (public-goods, ultimatum, time-preference tasks) against GPS and WVS benchmarks; the adapters’ construct direction should survive, the dispersion boundary should be measured again.
2. **OOS evaluation on non-WVS surveys:** AmericasBarometer (USA/MEX locally) and a second instrument; tests whether the construct bridge transfers beyond the 30-item map.

3. **New nomological restrictions with cvprofiles:** add rule-of-law, inequality (Gini), and institutional quality restrictions to the trust network; compute the construct-identified range $[L, U]$ for trust using the adapter composites as candidate measures — the partial-identification instrument makes the adapter-vs-human disagreement itself the measurement conclusion.
4. **Soft DPO / label-pruning ablation:** $|z| < 0.1$ threshold and preference-intensity weights on the frozen bank; expected to improve patience/risktaking direction, not dispersion.
5. **Country-named prompt as a documented baseline:** the persona run is now a committed artifact; report it in every table and figure as the literature-comparison anchor.

References

- [1] Muthukrishna, M., et al. (2020). Beyond Western, Educated, Industrial, Rich, and Democratic (WEIRD) psychology: Measuring and mapping scales of cultural and psychological distance. *Psychological Science*, 31(6), 678–701.
- [2] Falk, A., et al. (2018). Global evidence on economic preferences. *Quarterly Journal of Economics*, 133(4), 1645–1692.
- [3] Pfeifer, M., & Dalloul, A. (2026). Can LLMs mimic household surveys? From representative agents to population distributions. SSRN 6703798.
- [4] Bisbee, J., et al. (2024). Synthetic replacements for human survey data? The perils of large language models. *Political Analysis*.
- [5] Wu, Y., et al. (2024). Soft preference optimization: Aligning language models to expert preferences. ICML 2024.
- [6] Tao, Y., Viberg, O., Baker, R.S., & Kizilcec, R.F. (2024). Cultural bias and cultural alignment of large language models. *PNAS Nexus*, 3(9), pgae346.
- [7] Inglehart, R., & Welzel, C. (2005). *Modernization, Cultural Change, and Democracy*. Cambridge University Press.